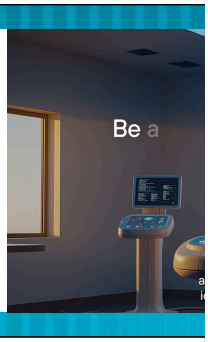
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Can AI replace junior workers?

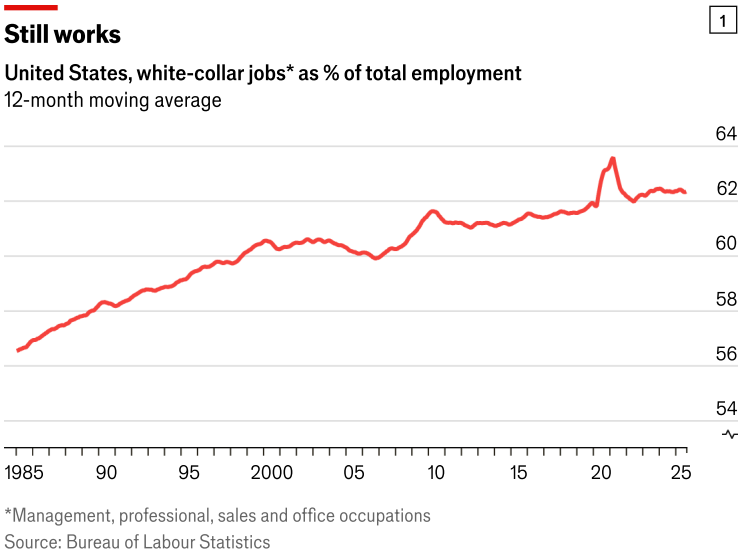
The effect is still hard to spot in official data. But a study of 300,000 companies suggests where hiring is weakening

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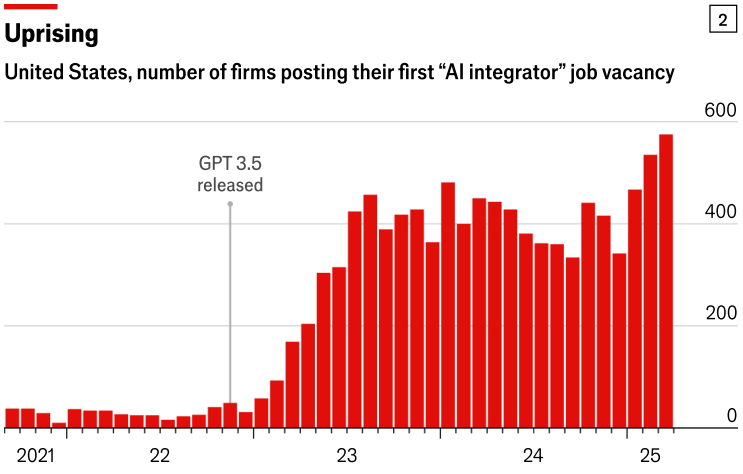
Oct 13th 2025 | 2 min read

America's economy is a puzzle. Growth remains healthy enough, yet just 22,000 jobs were added in August, down from 158,000 in April. Amid this lull, one worry is swirling: is generative AI starting to pinch human work?

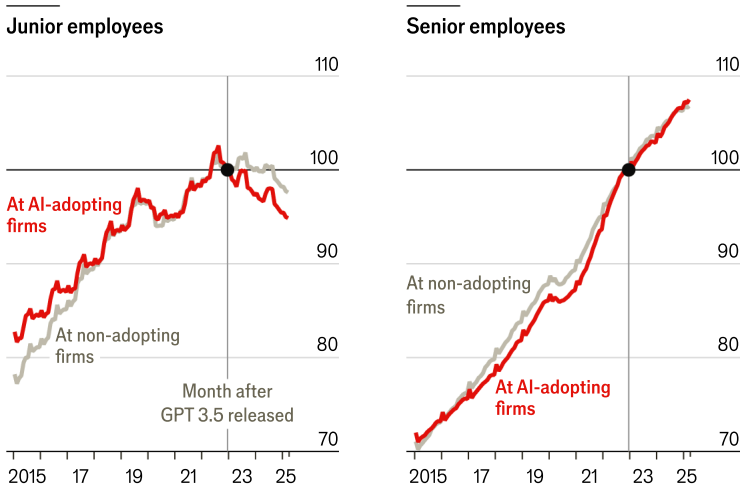
So far, an AI-driven jobs-pocalypse looks distant. Over the past year the share of white-collar jobs in total employment, the sort deemed most exposed to AI automation, has remained stable (see chart 1). Research by Yale University's Budget Lab, published in October, finds no large shift in the kinds of jobs people do since the debut of ChatGPT, a popular chatbot, in late 2022.



Data at the company level, though, show a subtle pattern emerging. New work by Seyed Hosseini and Guy Lichtinger, both doctoral students at Harvard University, tracks companies that hired "generative-AI integrators", whose jobs are to embed the technology into daily operations. Using AI to parse 200m job postings, they identified around 130,000 such vacancies at 10,600 firms that they called "AI adopters". Hiring of these sorts of workers picked up around the first quarter of 2023 (see chart 2), around the time when ChatGPT 3.5 was released. The other 274,000 companies in their database was a control group of sorts—those which did not hire staff to specifically overhaul their workflows with AI.

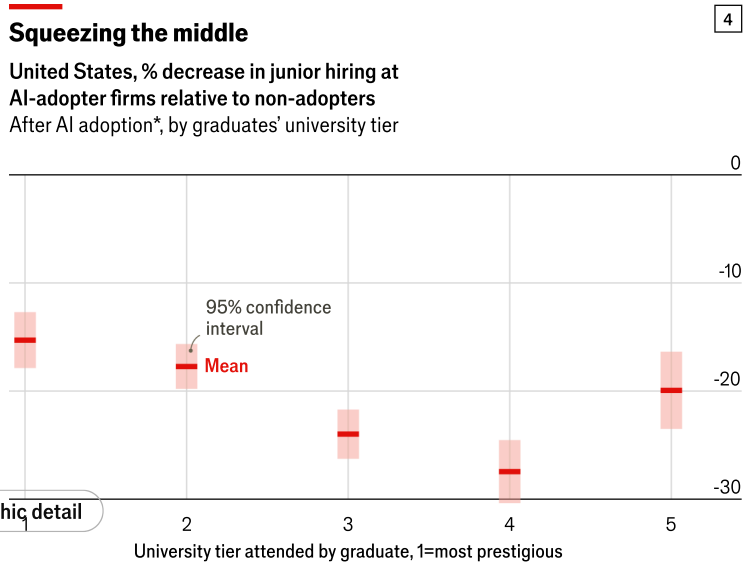


If AI had no bearing on employment, hiring at adopter firms might be expected to match that at non-adopters. Instead, the researchers found that while junior roles fell across the board after 2023, the decline over the next six quarters was 7.7% steeper at adopter firms (see chart 3). No such gap appeared in senior hiring. The kind of menial but mentally taxing work graduates do, such as debugging code or reviewing documents, seems especially easy to hand to machines. The fall, the authors note, came mainly from slower recruitment rather than layoffs.



Source: "Generative AI as seniority-biased technological change", by S.M. Hosseini & G. Lichtinger, SSRN working paper, 2025

Which graduates appeared to be squeezed? The study sorted universities into five tiers. Mid-tier graduates fared much worse than those from the top and bottom schools. The authors suspect firms may keep top-tier recruits for their specialist skills and lower-tier ones for their lower cost. The middle, they suggest, may face the strongest threat from AI.



*Firms adopted AI at different points in the sample from 2015-25

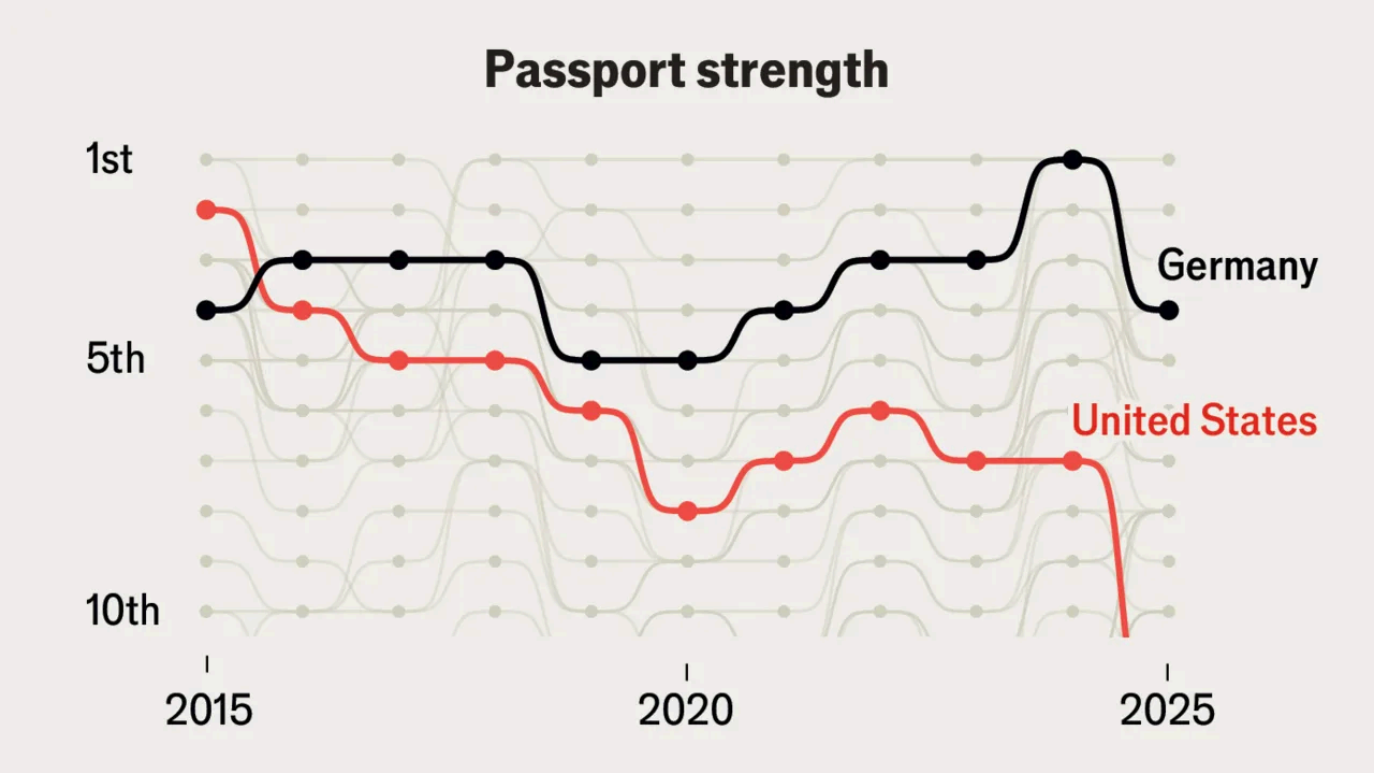
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Yet caution is warranted. For one thing only 17% of workers in the sample were employed by the AI adopters, meaning the opportunities for automation might be narrow. For another, hiring trends for junior staff have been volatile in recent years, disrupted by the covid-19 pandemic. If AI is a drag on graduate hiring, it is probably just one of several. ■

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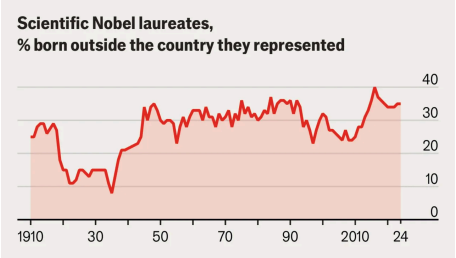


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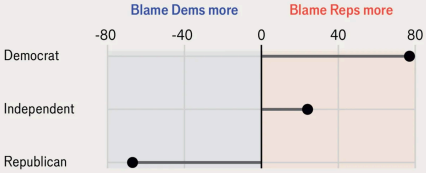
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